

Linear Algebra

Homework 2

Deadline: Sunday October 17th 2025 23:59

Write a system of equations that is equivalent to the given vector equation.

$$1. \quad x_1 \begin{bmatrix} -2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 8 \\ 5 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{cases} -2x_1 + 8x_2 + x_3 = 0 \\ 3x_1 + 5x_2 - 6x_3 = 0 \end{cases}$$

Write a vector equation that is equivalent to the given system of equations.

$$2. \quad \begin{aligned} 4x_1 + x_2 + 3x_3 &= 9 \\ x_1 - 7x_2 - 2x_3 &= 2 \\ 8x_1 + 6x_2 - 5x_3 &= 15 \end{aligned} \rightarrow x_1 \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ -7 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ -2 \\ -5 \end{bmatrix} = \begin{bmatrix} 9 \\ 2 \\ 15 \end{bmatrix}$$

Determine if **b** is a linear combination of **a**₁, **a**₂, and **a**₃.

$$3. \quad \mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & -5 \\ -2 & 5 & 0 & 11 \\ 2 & 5 & 8 & -7 \end{bmatrix} \xrightarrow{\substack{\times 2 \\ \times (-2)}} \begin{bmatrix} 1 & 0 & 2 & -5 \\ 0 & 5 & 4 & 1 \\ 0 & 5 & 4 & 3 \end{bmatrix} \Rightarrow \text{inconsistent} \Rightarrow \mathbf{b} \text{ is not a linear combination}$$

Determine if **b** is a linear combination of the vectors formed from the columns of the matrix **A**.

$$4. \quad A = \begin{bmatrix} 1 & -2 & -6 \\ 0 & 3 & 7 \\ 1 & -2 & 5 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 1 & -2 & 5 & 9 \end{bmatrix} \xrightarrow{\times (-1)} \begin{bmatrix} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 0 & 0 & 11 & -2 \end{bmatrix} \Rightarrow \begin{cases} x_1 = \frac{-5}{33} \\ x_2 = \frac{-41}{33} \\ x_3 = \frac{-2}{11} \end{cases}$$

$$5. \quad \text{Let } \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -5 \\ 1 \\ 7 \end{bmatrix}, \text{ and } \mathbf{y} = \begin{bmatrix} h \\ -1 \\ -5 \end{bmatrix}. \text{ For what value(s) of } h \text{ is } \mathbf{y} \text{ in the plane generated by } \mathbf{v}_1 \text{ and } \mathbf{v}_2? \rightarrow \begin{bmatrix} 1 & -5 & h \\ 0 & 1 & -1 \\ -4 & 7 & -5 \end{bmatrix} \xrightarrow{\times 4} \begin{bmatrix} 1 & -5 & h \\ 0 & 1 & -1 \\ 0 & -3 & -5+h \end{bmatrix} \xrightarrow{\times 3} \begin{bmatrix} 1 & -5 & h \\ 0 & 1 & -1 \\ 0 & -3 & -5+h \end{bmatrix} \xrightarrow{\times 3} \begin{bmatrix} 1 & -5 & h \\ 0 & 1 & -1 \\ 0 & 0 & -18+4h \end{bmatrix} \Rightarrow -18+4h=0, h=\frac{9}{2}$$

Use the definition of **Ax** to write the matrix equation as a vector equation or vice versa.

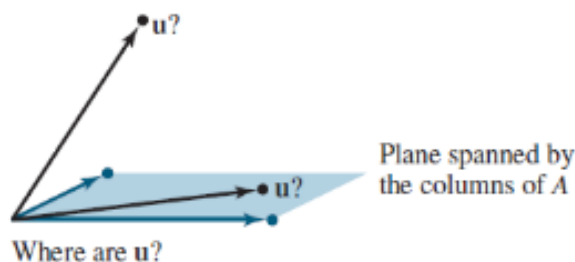
$$6. \quad \begin{bmatrix} 7 & -3 \\ 2 & 1 \\ 9 & -6 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ 12 \\ -4 \end{bmatrix} \rightarrow (-2) \begin{bmatrix} 7 \\ 2 \\ 9 \\ -3 \end{bmatrix} + (-5) \begin{bmatrix} -3 \\ 1 \\ -6 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ 12 \\ -4 \end{bmatrix}$$

$$7. \begin{cases} 4z_1 - 4z_2 - 5z_3 + 3z_4 = 4 \\ -2z_1 + 5z_2 + 4z_3 + 0z_4 = 13 \end{cases} \quad \begin{bmatrix} 4 & -4 & -5 & 3 \\ -2 & 5 & 4 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 13 \end{bmatrix}$$

$$7. \quad 4z_1 - 4z_2 - 5z_3 + 3z_4 = 4$$

$$8. \quad \text{Let } \mathbf{u} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} \text{ and } A = \begin{bmatrix} 3 & -5 \\ -2 & 6 \\ 1 & 1 \end{bmatrix}. \text{ Is } \mathbf{u} \text{ in the plane in } \mathbb{R}^3$$

spanned by the columns of A ? (See the figure.) Why or why not?



$$8. \quad x_1 \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 6 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$

$$\begin{cases} 3x_1 - 5x_2 = 0 \\ -2x_1 + 6x_2 = 4 \\ x_1 + x_2 = 4 \end{cases}$$

$$\begin{bmatrix} 3 & -5 & 0 \\ -2 & 6 & 4 \\ 1 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 4 \\ -2 & 6 & 4 \\ 3 & -5 & 0 \end{bmatrix} \xrightarrow{\times 2} \begin{bmatrix} 1 & 1 & 4 \\ -2 & 6 & 4 \\ 3 & -5 & 0 \end{bmatrix}$$

$$9. \quad \text{Let } \mathbf{u} = \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix} \text{ and } A = \begin{bmatrix} 5 & 8 & 7 \\ 0 & 1 & -1 \\ 1 & 3 & 0 \end{bmatrix}. \text{ Is } \mathbf{u} \text{ in the subset of } \mathbb{R}^3 \text{ spanned by the columns of } A? \text{ Why or why not?}$$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 8 & 12 \\ 0 & -8 & -12 \end{bmatrix} \xrightarrow{\times 2} \begin{bmatrix} 1 & 1 & 4 \\ 0 & 8 & 12 \\ 0 & -8 & -12 \end{bmatrix} \xrightarrow{\times 2} \begin{bmatrix} 1 & 1 & 4 \\ 0 & 8 & 12 \\ 0 & -8 & -12 \end{bmatrix}$$

$$10. \quad \text{Let } \mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ -3 \\ 8 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 4 \\ -1 \\ -5 \end{bmatrix}.$$

Does $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ span $\mathbb{R}^3$? Why or why not?

Yes, the equation is consistent.

$$9. \quad \begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 1 & 3 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & -1 & -3 \\ 5 & 8 & 7 & 2 \end{bmatrix} \xrightarrow{\times 5} \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & -1 & -3 \\ 5 & 8 & 7 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & -7 & 7 & -8 \end{bmatrix} \xrightarrow{\times 7} \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & -29 \end{bmatrix}$$

No, the equation is inconsistent.

10.

$$\begin{bmatrix} 0 & 0 & 4 \\ 0 & -3 & -1 \\ -2 & 8 & -5 \end{bmatrix}$$

$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ doesn't span $\mathbb{R}^3$

$\Rightarrow$ inconsistent